

SR-MPLS L3VPN Engineering Poster

A Comprehensive Technical Reference for Service Provider Networks

SECTION 1: EXECUTIVE HEADER

Lab Title

L3VPN BASIC ON SEGMENT ROUTING

Layer 3 VPN on Segment Routing Infrastructure

Key Metadata

Attribute	Value
Technology Stack	SR-MPLS, OSPF, MP-BGP VPNv4
Vendor	Cisco IOS-XR 6.3.1
Difficulty Level	Advanced (CCIE/JNCIE)
Estimated Completion Time	2–3 hours
Prerequisites	OSPF, BGP, MPLS fundamentals
Lab Version	1.0
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Lab Objectives

This lab exercise enables engineers to understand and implement Layer 3 VPN services over a Segment Routing MPLS infrastructure. The specific learning outcomes include:

- Configure SR-MPLS Underlay:** Establish OSPF-based routing with Segment Routing enabled, including Prefix-SID assignment and SRGB configuration on all core routers.
- Deploy MP-BGP VPNv4 Overlay:** Establish multiprotocol BGP sessions between Provider Edge routers to advertise customer VRF prefixes with proper route distinguishers and route targets.
- Implement VRF-Based Customer Isolation:** Create separate virtual routing and forwarding instances for different customer groups (Accounting and HR) with strict traffic isolation.

4. **Verify End-to-End L3VPN Connectivity:** Validate that customer edge routers can reach remote sites through the SR-MPLS backbone, including verification of label stacking and data plane forwarding.
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SECTION 2: DETAILED NETWORK TOPOLOGY

Topology Overview

The lab topology consists of eight routers organized into three functional groups:

Provider Edge (PE) Routers: R1 (XRV1) and R4 (XRV4) serve as the entry and exit points for customer traffic. These routers run OSPF, Segment Routing, BGP VPNv4, and maintain customer VRFs.

Provider (P) Routers: R2 (XRV2) and R3 (XRV3) function as transit nodes in the MPLS backbone. They participate in OSPF and Segment Routing but do not require BGP or VRF configuration.

Customer Edge (CE) Routers: R5, R6, R7, and R8 represent customer sites. R5 and R6 connect to the left PE (R1), while R7 and R8 connect to the right PE (R4). These routers use static routing and default routes to reach remote sites.

Core Backbone Links

The core backbone forms a linear path from R1 to R4 through R2 and R3:

- **R1 ↔ R2:** Interface Gi0/0/0/0 on both sides, subnet 10.1.12.0/24, configured as OSPF point-to-point
- **R2 ↔ R3:** Interface Gi0/0/0/1 on both sides, subnet 10.1.23.0/24, configured as OSPF point-to-point
- **R3 ↔ R4:** Interface Gi0/0/0/2 on both sides, subnet 10.1.34.0/24, configured as OSPF point-to-point

Customer Attachment Points

Left PE (R1) Attachments:

- R5 (Accounting customer) connects via Gi0/0/0/1 in VRF Accounting, subnet 192.168.15.0/24
- R6 (HR customer) connects via Gi0/0/0/2 in VRF HR, subnet 192.168.16.0/24

Right PE (R4) Attachments:

- R7 (Accounting customer) connects via Gi0/0/0/0 in VRF Accounting, subnet 192.168.47.0/24
- R8 (HR customer) connects via Gi0/0/0/1 in VRF HR, subnet 192.168.48.0/24

Segment Routing Identifiers

Each router is assigned a unique Prefix-SID index used for label calculation:

Router	Loopback	Prefix-SID Index	Calculated Label (SRGB 16000–23999)
R1	1.1.1.1/32	1	16001
R2	2.2.2.2/32	2	16002
R3	3.3.3.3/32	3	16003
R4	4.4.4.4/32	4	16004

SECTION 3: DEVICE INVENTORY MATRIX

Node	Vendor	Role	Loopback	Router ID	Prefix-SID Index	SRGB Range	Protocols Enabled
R1 (XRV1)	Cisco IOS-XR	Provider Edge	1.1.1.1/32	1.1.1.1	1	16000–23999	OSPF, SR-MPLS, BGP VPNv4
R2 (XRV2)	Cisco IOS-XR	Provider (Transit)	2.2.2.2/32	2.2.2.2	2	16000–23999	OSPF, SR-MPLS
R3 (XRV3)	Cisco IOS-XR	Provider (Transit)	3.3.3.3/32	3.3.3.3	3	16000–23999	OSPF, SR-MPLS
R4 (XRV4)	Cisco IOS-XR	Provider Edge	4.4.4.4/32	4.4.4.4	4	16000–23999	OSPF, SR-MPLS, BGP VPNv4

SECTION 4: COMPLETE IP ADDRESSING PLAN

Loopback Addresses

Loopback interfaces serve as router identifiers and are used as OSPF router IDs and BGP update sources:

Router	Interface	IP Address	Prefix-SID Index	Label
R1	Loopback0	1.1.1. ¹ / ₃₂	1	16001
R2	Loopback0	2.2.2. ² / ₃₂	2	16002
R3	Loopback0	3.3.3. ³ / ₃₂	3	16003
R4	Loopback0	4.4.4. ⁴ / ₃₂	4	16004

Core Backbone Links

These point-to-point links form the OSPF Area 0 underlay and carry MPLS traffic:

Link	Router A	Interface A	IP A	Router B	Interface B	IP B	Subnet
1–2	R1	Gi0/0/0/0	10.1.12.1	R2	Gi0/0/0/0	10.1.12.2	10.1.12.0/24
2–3	R2	Gi0/0/0/1	10.1.23.2	R3	Gi0/0/0/1	10.1.23.3	10.1.23.0/24
3–4	R3	Gi0/0/0/2	10.1.34.3	R4	Gi0/0/0/2	10.1.34.4	10.1.34.0/24

VRF Interface Addressing

VRF interfaces on PE routers connect to customer edge routers and are configured with customer-facing subnets:

Router	VRF	Interface	IP Address	RD	RT Import	RT Export
R1	Accounting	Gi0/0/0/1	192.168.15. $\frac{1}{24}$	1:1	1:100	1:100
R1	HR	Gi0/0/0/2	192.168.16. $\frac{1}{24}$	2:2	1:200	1:200
R4	Accounting	Gi0/0/0/0	192.168.47. $\frac{1}{24}$	auto	1:100	1:100
R4	HR	Gi0/0/0/1	192.168.48. $\frac{1}{24}$	auto	1:200	1:200

Customer Edge Interface Addressing

Customer routers use simple IP addressing with static default routes pointing to the PE:

CE Router	VRF	Interface	IP Address	Gateway (PE)	Default Route
R5	Accounting	Eth0/0	192.168.15. $\frac{5}{24}$	192.168.15.1	0.0.0.0/0 → 192.168.15.1
R6	HR	Eth0/0	192.168.16. $\frac{6}{24}$	192.168.16.1	0.0.0.0/0 → 192.168.16.1
R7	Accounting	Eth0/0	192.168.47. $\frac{7}{24}$	192.168.47.1	0.0.0.0/0 → 192.168.47.1
R8	HR	Eth0/0	192.168.48. $\frac{8}{24}$	192.168.48.1	0.0.0.0/0 → 192.168.48.1

SECTION 5: PROTOCOL ARCHITECTURE

Control Plane Stack

The control plane consists of three layered protocols working in concert:

Layer 1 – Interior Gateway Protocol (OSPF)

OSPF discovers network topology and distributes reachability information. In this lab, OSPF operates in Area 0 with all routers configured as router ID from their loopback addresses. OSPF is responsible for advertising loopback prefixes with Prefix-SID TLVs, which enable Segment Routing label calculation.

Layer 2 – Segment Routing MPLS

Segment Routing integrates label distribution directly into OSPF, eliminating the need for a separate Label Distribution Protocol (LDP). Each router calculates labels for all loopback prefixes using the formula: Label = SRGB Base + Prefix-SID Index. For example, R1 with Prefix-SID index 1 and SRGB 16000–23999 calculates label 16001 for its loopback 1.1.1.¹/₃₂.

Layer 3 – Multiprotocol BGP (MP-BGP)

MP-BGP establishes iBGP sessions between PE routers using their loopback addresses as update sources. The VPNv4 address family enables advertisement of customer prefixes with route distinguishers (RD) and route targets (RT), allowing VRF-based prefix import/export control.

Data Plane Stack

Packets in SR-MPLS L3VPN carry a two-label stack:

Transport Label (Top Label)

The transport label is the Prefix-SID of the egress PE router. For example, when R1 sends traffic destined for a prefix advertised by R4, R1 pushes label 16004 (R4's Prefix-SID) as the top label. This label guides the packet through the SR-MPLS underlay to R4's loopback.

VPN Label (Bottom Label)

The VPN label is allocated by MP-BGP for each VRF prefix. For example, the Accounting VRF prefix 192.168.47.0/24 advertised by R4 receives label 24003. This label is used by the egress PE (R4) to determine which VRF to forward the packet into after the transport label is popped.

Protocol Relationships and Dependencies

The three layers form a dependency chain: OSPF discovers routes and distributes Prefix-SID information → SR-MPLS calculates labels from SRGB + Prefix-SID index → MP-BGP advertises VRF prefixes with VPN labels. This three-layer model separates concerns: the underlay (OSPF + SR) handles transport, the overlay (MP-BGP) handles service routing, and VRFs handle customer isolation.

SECTION 6: NODE CONFIGURATION HIGHLIGHTS

R1 (XRV1) – Provider Edge Configuration

OSPF and Segment Routing Configuration

```
router ospf 1
  router-id 1.1.1.1
  segment-routing mpls
  segment-routing forwarding mpls
  area 0
    interface Loopback0
      passive enable
      prefix-sid index 1
    !
    interface GigabitEthernet0/0/0/0
      network point-to-point
    !
  !
!

segment-routing
  global-block 16000 23999
!
```

BGP VPNv4 and VRF Configuration

```
router bgp 12345
  bgp router-id 1.1.1.1
  address-family vpnv4 unicast
  !
  neighbor-group L3VPN
    remote-as 12345
    update-source Loopback0
    address-family vpnv4 unicast
  !
  !
  neighbor 4.4.4.4
    use neighbor-group L3VPN
  !
  vrf Accounting
    rd 1:1
    address-family ipv4 unicast
      network 192.168.15.0/24
    !
  !
  vrf HR
    rd 2:2
    address-family ipv4 unicast
      network 192.168.16.0/24
    !
  !
  !

vrf Accounting
  address-family ipv4 unicast
    import route-target
      1:100
    !
    export route-target
      1:100
    !
  !
  !

vrf HR
  address-family ipv4 unicast
    import route-target
      1:200
    !
    export route-target
      1:200
```



```
!  
!  
!
```

R2 (XRV2) – Provider (Transit) Configuration

OSPF and Segment Routing Configuration

```
router ospf 1  
  router-id 2.2.2.2  
  segment-routing mpls  
  segment-routing forwarding mpls  
  area 0  
    interface Loopback0  
      passive enable  
      prefix-sid index 2  
    !  
    interface GigabitEthernet0/0/0/0  
      network point-to-point  
    !  
    interface GigabitEthernet0/0/0/1  
      network point-to-point  
    !  
  !  
  !  
  !  
  
segment-routing  
  global-block 16000 23999  
!
```

Note: P nodes do not require BGP or VRF configuration. They forward packets based solely on transport labels in the MPLS forwarding table.

R3 (XRV3) – Provider (Transit) Configuration

R3 configuration is identical to R2, with the only difference being the Prefix-SID index (3 instead of 2) and the interface numbers (Gi0/0/0/1 and Gi0/0/0/2 instead of Gi0/0/0/0 and Gi0/0/0/1).

R4 (XRV4) – Provider Edge Configuration

OSPF and Segment Routing Configuration

```
router ospf 1
  router-id 4.4.4.4
  segment-routing mpls
  segment-routing forwarding mpls
  area 0
    interface Loopback0
      passive enable
      prefix-sid index 4
    !
    interface GigabitEthernet0/0/0/2
      network point-to-point
    !
  !
!

segment-routing
  global-block 16000 23999
!
```

BGP VPNv4 and VRF Configuration

```
router bgp 12345
  bgp router-id 4.4.4.4
  address-family vpnv4 unicast
  !
  neighbor-group L3VPN
    remote-as 12345
    update-source Loopback0
    address-family vpnv4 unicast
    !
  !
  neighbor 1.1.1.1
    use neighbor-group L3VPN
    !
  vrf Accounting
    rd auto
    address-family ipv4 unicast
      network 192.168.47.0/24
    !
  !
  vrf HR
    rd auto
    address-family ipv4 unicast
      network 192.168.48.0/24
    !
  !
  !

vrf Accounting
  address-family ipv4 unicast
    import route-target
      1:100
    !
    export route-target
      1:100
    !
  !
  !

vrf HR
  address-family ipv4 unicast
    import route-target
      1:200
    !
    export route-target
      1:200
```

!
!
!

SECTION 7: DEPLOYMENT WORKFLOW

The configuration of SR-MPLS L3VPN follows a logical six-step sequence. Each step builds upon the previous one, ensuring that the underlay is stable before the overlay is deployed.

Step 1: Configure Hostnames and IP Addressing

Begin by setting hostnames on all routers and configuring IP addresses on all interfaces. Configure loopback interfaces with /32 addresses (1.1.1.1, 2.2.2.2, 3.3.3.3, 4.4.4.4). Configure core backbone links with 10.1.x.x/24 subnets. On PE routers, configure VRF interfaces with customer-facing subnets (192.168.x.x/24). Verify basic IP connectivity with ping across all links.

Step 2: Configure Segment Routing Global Block

Define the SRGB on each router using the command `segment-routing global-block 16000 23999`. The SRGB defines the label range used for Prefix-SID calculations. All routers in the domain should use the same SRGB to ensure consistent label allocation.

Step 3: Enable OSPF with Segment Routing

Configure OSPF process 1 on all routers with router IDs matching loopback addresses. Enable Segment Routing MPLS with the commands `segment-routing mpls` and `segment-routing forwarding mpls` within the OSPF process. Assign Prefix-SID indices to loopback interfaces (1, 2, 3, 4 respectively). Configure core links as point-to-point to avoid DR/BDR election and ensure proper label calculation. Verify OSPF adjacencies reach FULL state and that MPLS labels appear in the forwarding table.

Step 4: Configure VRFs on PE Routers

Create two VRFs on PE routers: Accounting (RD 1:1, RT 1:100) and HR (RD 2:2, RT 1:200). Assign customer-facing interfaces to appropriate VRFs. Verify VRF configuration with `show vrf` and `show interface vrf` commands.

Step 5: Configure BGP VPNv4 Between PEs

Establish an iBGP session between R1 and R4 using ASN 12345. Configure the neighbor relationship with `update-source Loopback0` to ensure the session uses stable loopback IPs. Enable the VPNv4 address family and activate the neighbor in that AF. Advertise VRF prefixes using the `network` command within each VRF's address family. Verify BGP session is established and routes are advertised with `show bgp vpnv4 unicast`.

Step 6: Configure CE Routers and Verify

Configure CE routers (R5–R8) with IP addresses and default routes pointing to the PE. Verify end-to-end connectivity with ping from CE to remote CE. Use traceroute to verify label stacking and confirm

that packets traverse the correct path through the backbone.

SECTION 8: VERIFICATION COMMANDS AND EXPECTED OUTPUTS

OSPF Verification

Command: `show ospf neighbor`

Expected Output: All neighbors (R2, R3, R4) should be in FULL state. This indicates that OSPF adjacencies are established and the topology database is synchronized.

Command: `show ospf database`

Expected Output: The OSPF database should contain router link state advertisements (LSAs) from all four routers. Each LSA should include the Prefix-SID TLV for the loopback prefix.

MPLS Verification

Command: `show mpls forwarding`

Expected Output: The MPLS forwarding table should display Prefix-SID labels for all loopback prefixes:

- 16002 (R2) with action Pop (penultimate hop popping)
- 16003 (R3) with action Swap to 16003
- 16004 (R4) with action Swap to 16004

This output confirms that SR-MPLS is calculating and installing labels correctly.

Command: `show segment-routing mpls state`

Expected Output: Displays the SRGB range (16000–23999), Prefix-SID index, calculated label, and status (Active).

BGP Verification

Command: `show bgp vpnv4 unicast`

Expected Output: Displays VPNv4 routes from both local and remote PEs. For each VRF, the output should show:

- Local routes (next hop 0.0.0.0) with weight 32768
- Remote routes (next hop 4.4.4.4 for R1, or 1.1.1.1 for R4) with local preference 100

Command: `show bgp vrf Accounting`

Expected Output: Displays prefixes for the Accounting VRF only. Should show local prefix

192.168.15.0/24 and remote prefix 192.168.47.0/24.

End-to-End Connectivity Verification

Command from R5: `ping 192.168.47.7`

Expected Output: 100% success rate with round-trip times in the range of 30–200 ms, depending on network load.

Command from R5: `traceroute 192.168.47.7`

Expected Output: Displays the path through the backbone with MPLS label information. The output should show:

- Hop 1: 192.168.15.1 (R1 PE)
- Hop 2: 10.1.12.2 (R2 P) with label stack [²⁴⁰⁰⁴/₂₄₀₀₃]
- Hop 3: 10.1.23.3 (R3 P) with label stack [²⁴⁰⁰⁴/₂₄₀₀₃]
- Hop 4: 10.1.34.4 (R4 PE)
- Hop 5: 192.168.47.7 (R7 CE)

The label stack [²⁴⁰⁰⁴/₂₄₀₀₃] indicates that the transport label is 24004 and the VPN label is 24003.

SECTION 9: TROUBLESHOOTING MATRIX

Issue	Symptoms	Verification Command	Root Cause	Fix
OSPF Down	Neighbors in EXSTART/EXCHANGE state	<code>show ospf neighbor</code>	Interface MTU mismatch or network type misconfiguration	Verify MTU=1500, set network point-to-point on core links
SR Labels Missing	No Prefix-SID labels in MPLS table	<code>show mpls forwarding</code>	SR-MPLS not enabled in OSPF or Prefix-SID index not set	Enable segment-routing mpls in OSPF, set prefix-sid index on Lo0
BGP Down	Neighbor in Idle or Connect state	<code>show bgp summary</code>	Update-source not set to Loopback0 or ASN mismatch	Set update-source Loopback0 , verify ASN 12345 on both PEs
VRF Routes Not Advertised	<code>show bgp vpnv4</code> shows only local routes	<code>show bgp vpnv4 unicast</code>	VRF not configured on PE or RT mismatch	Verify VRF exists, check RT import/export match between PEs
CE Ping Fails	100% packet loss from CE to remote CE	<code>traceroute</code> from CE	VPN label not allocated or transport label issue	Verify <code>show mpls forwarding</code> shows VPN labels, check BGP routes

Issue	Symptoms	Verification Command	Root Cause	Fix
Wrong VRF Reachability	Accounting CE can reach HR CE (isolation broken)	<code>show bgp vrf [VRF-name]</code>	RT values incorrect or overlapping	Verify RT 1:100 for Accounting, RT 1:200 for HR

SECTION 10: LEARNING OUTCOMES AND REAL-WORLD APPLICATION

What We Learned

This lab demonstrates the complete architecture of SR-MPLS L3VPN, a modern service provider technology that combines the simplicity of Segment Routing with the scalability of MPLS-based VPN services. Engineers completing this lab gain hands-on experience with three critical concepts: (1) SR-MPLS simplifies the control plane by encoding labels directly in the IGP, eliminating the need for LDP; (2) Prefix-SID provides consistent, globally unique labels across the entire domain; (3) MP-BGP VPNv4 enables multi-tenant services with strict customer isolation through VRF-based routing.

Why It Matters

SR-MPLS L3VPN addresses key operational challenges in modern networks. First, it reduces control plane complexity compared to traditional MPLS with LDP, which requires managing two separate protocols. Second, it enables 100% fast-reroute coverage through TI-LFA (Topology Independent Loop-Free Alternates), providing automatic protection for all prefixes without requiring RSVP-TE. Third, it scales to thousands of services per PE router, making it ideal for large service providers. Fourth, it serves as the foundation for emerging technologies like segment routing policies and SD-WAN. Finally, it is the industry standard for modern ISPs, telcos, and large enterprises.

Real-World Applications

Service Provider L3VPN: ISPs offer managed VPN services to enterprise customers, guaranteeing service level agreements (SLAs) for latency, jitter, and availability. SR-MPLS L3VPN provides the scalable, resilient infrastructure to support thousands of such services.

Data Center Interconnect: Large enterprises operate multiple data centers and require low-latency, high-bandwidth connectivity between them. SR-MPLS L3VPN enables deterministic routing and fast failover, ensuring business continuity.

5G Backhaul: Mobile operators use SR-MPLS L3VPN to transport RAN (Radio Access Network) traffic from cell sites to core network elements. The low latency and guaranteed bandwidth are

critical for 5G service quality.

SD-WAN Underlay: SD-WAN solutions overlay application-aware routing on top of underlay networks. SR-MPLS provides the underlay transport with QoS guarantees and fast convergence.

Certification Topics Covered

CCIE Service Provider: This lab covers multiple CCIE SP exam topics including SR-MPLS configuration, L3VPN design and deployment, BGP VPNv4, and OSPF with Segment Routing.

JNCIE-SP: Juniper's equivalent certification covers similar topics with Junos OS syntax.

HCIA/HCIP Routing: Huawei's certification program includes MPLS fundamentals and VPN services, which this lab reinforces.

SECTION 11: INTERVIEW PREPARATION – 30 KEY QUESTIONS

Conceptual Questions

Q1: What is Segment Routing and how does it differ from traditional MPLS?

A: SR encodes path information directly in packet headers as segment identifiers (SIDs). Unlike traditional MPLS which uses LDP to distribute labels hop-by-hop, SR calculates labels from a local SRGB using a global index. This eliminates the need for LDP, reducing control plane complexity and enabling faster convergence.

Q2: Explain the concept of Prefix-SID and how it relates to SRGB.

A: A Prefix-SID is a global index assigned to a loopback prefix (e.g., index 1 for 1.1.1.1). Each router calculates the actual label by adding the index to its SRGB base. For example, with SRGB 16000–23999, index 1 = label 16001. This ensures consistent labels across the domain.

Q3: What is the role of MP-BGP VPNv4 in SR-MPLS L3VPN?

A: MP-BGP VPNv4 advertises customer prefixes between PE routers with route distinguishers (RD) and route targets (RT). It allocates VPN labels for each prefix and ensures only authorized VRFs receive the routes. The VPN label is stacked below the transport label for end-to-end service delivery.

Q4: Describe the label stack in SR-MPLS L3VPN. What are the two labels and their purposes?

A: The label stack has two labels: (1) Transport Label (top) – Prefix-SID of the egress PE, used for routing through the SR-MPLS underlay. (2) VPN Label (bottom) – allocated by MP-BGP, used by the egress PE to identify which VRF to forward to. Example: [¹⁶⁰⁰⁴/₂₄₀₀₃] means transport to R4, then use label 24003 for VRF lookup.

Q5: What is the significance of Route Targets in L3VPN?

A: Route Targets (RT) control which VRFs import/export prefixes. A prefix with RT 1:100 is only imported by VRFs configured to import 1:100. This enables multi-tenant isolation – Accounting VRF

(RT 1:100) cannot see HR VRF (RT 1:200) routes, ensuring customer separation.

Q6: Explain the difference between a P node and a PE node in this topology.

A: P (Provider) nodes (R2, R3) only forward based on transport labels – they have no VRF or BGP configuration. PE (Provider Edge) nodes (R1, R4) terminate customer connections, run BGP VPNv4, and manage VRFs. PEs add/remove labels; P nodes only swap them.

Q7: What is Penultimate Hop Popping (PHP) and when is it used?

A: PHP is when the penultimate hop (P node) pops the transport label before forwarding to the egress PE. This reduces PE processing. However, modern SR designs often use Uniform Propagation or Explicit Null to maintain full label stacks for OAM visibility and troubleshooting.

Q8: How does SR-MPLS enable 100% fast-reroute coverage?

A: SR-MPLS enables TI-LFA (Topology Independent Loop-Free Alternates) which pre-computes backup paths for every link/node failure. Unlike traditional LDP which requires RSVP-TE for FRR, SR-MPLS provides FRR natively for all prefixes without additional protocols.

Configuration Questions

Q9: What commands enable SR-MPLS in OSPF?

A: In OSPF process: `segment-routing mpls` and `segment-routing forwarding mpls`. On Loopback0: `prefix-sid index [value]`. These commands enable SR-MPLS and assign a Prefix-SID index to the router's loopback.

Q10: How do you configure the SRGB on a router?

A: Under `segment-routing` global config: `global-block 16000 23999`. This defines the label range for Prefix-SID calculations. All routers in the domain should use the same SRGB for consistent label allocation.

Q11: What is the purpose of setting network point-to-point on core links?

A: This tells OSPF to treat the link as point-to-point (no DR/BDR election) and enables faster adjacency formation. It also ensures proper Prefix-SID label calculation for the link. Without it, OSPF may elect a DR and delay convergence.

Q12: How do you configure a VRF and associate it with an interface?

A: Create VRF: `vrf [name] → address-family ipv4 unicast → import/export route-target [RT]`. Assign interface: `interface [name] → vrf [name] → ipv4 address [IP]`. The interface now belongs to that VRF and only sees routes matching the RT.

Q13: What is the difference between RD and RT in VPNv4?

A: RD (Route Distinguisher) makes VPN prefixes globally unique by prepending it to the prefix (e.g., 1:1:192.168.15.0/24). RT (Route Target) is a policy tool that controls import/export between VRFs. RD is per-VRF; RT can be shared across VRFs for route leaking.

Q14: How do you establish an iBGP session between PE routers?

A: Configure BGP process with same ASN on both PEs. Create neighbor with remote-as matching local ASN. Set `update-source Loopback0` to ensure the session uses stable loopback IPs. Enable `address-family vpnv4 unicast` and activate the neighbor in that AF.

Q15: What commands advertise a VRF prefix in BGP?

A: Under `router bgp` → `vrf [name]` → `address-family ipv4 unicast` → `network [prefix] [mask]`. This advertises the prefix with the VRF's RD and RT, making it available for import by remote PEs with matching RT.

Troubleshooting Questions

Q16: OSPF neighbors are stuck in EXSTART state. What could be wrong?

A: Likely causes: (1) MTU mismatch – verify `show ip interface [int]` MTU=1500. (2) Network type misconfiguration – set `network point-to-point`. (3) Authentication failure – check OSPF auth config. (4) Subnet mismatch – verify IPs are in same /24.

Q17: show mpls forwarding shows no Prefix-SID labels. What's the issue?

A: SR-MPLS not enabled. Check: (1) `show ospf` – verify `segment-routing mpls` is active. (2) `show segment-routing` – verify SRGB is configured. (3) `show ospf database` – verify loopback prefixes have Prefix-SID TLVs. Fix: enable SR-MPLS and set prefix-sid index on Lo0.

Q18: BGP neighbors are in Idle state. How do you troubleshoot?

A: (1) `show bgp summary` – verify neighbor IP and ASN. (2) `show route` – verify loopback reachability (update-source). (3) `show tcp brief` – check if TCP port 179 is open. (4) Check for ACLs blocking BGP. (5) Verify `update-source Loopback0` is set.

Q19: CE can ping local PE but not remote CE. What's the issue?

A: (1) Check `show bgp vpnv4` – remote VRF prefix not advertised. (2) Verify RT match between PEs. (3) Check `show mpls forwarding` – VPN labels present? (4) Verify transport label reachability to remote PE. (5) Check CE default route points to correct PE interface.

Q20: Accounting CE can reach HR CE (VRF isolation broken). What happened?

A: (1) RT misconfiguration – Accounting and HR have same RT. (2) VRF interface assigned to wrong VRF. (3) BGP advertising wrong RD. Fix: verify `show vrf [name]` RT values are distinct (1:100 vs 1:200). Check interface VRF assignment.

Design Questions

Q21: Why is SR-MPLS preferred over traditional LDP for modern networks?

A: (1) Simpler control plane – no separate LDP protocol. (2) Faster convergence – labels calculated from IGP. (3) 100% FRR coverage via TI-LFA. (4) Better scalability – no per-prefix label distribution. (5) Foundation for segment routing policies and SD-WAN.

Q22: How would you scale this lab to support 1000 customers?

A: (1) Add more PE routers for load distribution. (2) Create one VRF per customer or use VRF lite with RT filtering. (3) Use BGP route reflectors to reduce full-mesh BGP sessions. (4) Implement hierarchical OSPF with multiple areas. (5) Use segment routing policies for traffic engineering.

Q23: What changes would you make to support inter-AS L3VPN?

A: (1) Use Option C (BGP-LU) between ASes – each AS advertises routes with labels. (2) Configure eBGP between ASBRs. (3) Use different ASNs for each AS. (4) Maintain separate SRGB ranges per AS. (5) Ensure RT consistency across ASes for route import.

Q24: How would you implement QoS for this L3VPN service?

A: (1) Use MPLS EXP bits in label stack for QoS marking. (2) Configure ingress policing on PE CE-facing interfaces. (3) Use traffic classes in OSPF for link priorities. (4) Implement per-VRF queue policies. (5) Mark traffic at ingress, honor markings at egress.

Q25: What monitoring and telemetry would you implement for this network?

A: (1) SNMP for OSPF/BGP neighbor status. (2) NetFlow/sFlow for traffic analysis. (3) BFD for fast failure detection. (4) Syslog for event logging. (5) Model-driven telemetry (gNMI) for real-time metrics. (6) LSPA for label verification.

Q26: How does this SR-MPLS L3VPN architecture support 5G backhaul?

A: (1) Low latency – direct MPLS forwarding without complex routing. (2) Guaranteed bandwidth – per-VRF QoS policies. (3) Fast convergence – SR-MPLS reacts quickly to failures. (4) Scalability – supports thousands of RAN sites as VRFs. (5) Deterministic paths – segment routing policies enforce specific routes.

Q27: What are the advantages of using auto RD on R4 compared to static RD on R1?

A: Auto RD simplifies configuration by automatically generating a unique RD based on the router's loopback and VRF name. This is useful in large deployments where manually assigning RDs to hundreds of VRFs is error-prone. However, static RDs provide explicit control and are preferred in smaller, well-planned networks.

Q28: How would you implement redundancy for the BGP session between R1 and R4?

A: (1) Add a second physical link between R1 and R4 with a different subnet. (2) Configure BGP multihop to allow the session to survive a single link failure. (3) Use BFD for fast failure detection. (4) Implement OSPF multi-area design with redundant paths. (5) Consider using BGP route reflectors for session resilience.

Q29: What is the impact of changing the SRGB range on an operational network?

A: Changing the SRGB range requires careful planning as it affects all Prefix-SID label calculations. All routers must be updated simultaneously to avoid label conflicts. During the transition, traffic may be disrupted if labels change. It is recommended to plan SRGB changes during maintenance windows and to use graceful restart mechanisms to minimize impact.

Q30: How would you migrate from traditional LDP-based MPLS to SR-MPLS L3VPN?

A: (1) Deploy SR-MPLS alongside LDP in a dual-stack mode. (2) Gradually shift traffic from LDP to SR-MPLS using traffic engineering policies. (3) Monitor both stacks for stability and performance. (4) Once all traffic is on SR-MPLS, decommission LDP. (5) Perform thorough testing at each stage to ensure service continuity.

SECTION 12: KEY TAKEAWAYS AND SUMMARY

Architecture Overview

This lab demonstrates a three-layer architecture that separates concerns and enables scalable multi-tenant services. The **underlay layer** consists of OSPF and Segment Routing, which together provide a simplified, scalable transport infrastructure. The **overlay layer** uses MP-BGP VPNv4 to advertise customer prefixes with proper route distinguishers and route targets. The **service layer** uses VRFs to isolate customer traffic and enforce routing policies.

Protocol Relationships

OSPF discovers network topology and distributes Prefix-SID information. Segment Routing converts this information into labels using a simple formula: $\text{Label} = \text{SRGB Base} + \text{Prefix-SID Index}$. MP-BGP advertises customer prefixes with VPN labels, enabling the egress PE to forward packets into the correct VRF. The combination creates a scalable, resilient platform for multi-tenant services without requiring a separate label distribution protocol.

Control Plane Summary

The control plane is simplified compared to traditional MPLS. Instead of running OSPF for IGP, LDP for label distribution, and BGP for VPN routing, SR-MPLS L3VPN uses OSPF (which now includes label distribution via Prefix-SID TLVs) and BGP (for VPN routing). This reduction in protocols decreases operational complexity and improves convergence speed.

Data Plane Summary

Packets carry a two-label stack: the transport label (Prefix-SID of the egress PE) and the VPN label (allocated by MP-BGP for the VRF prefix). The transport label guides the packet through the SR-MPLS underlay to the egress PE. The VPN label is used by the egress PE to determine which VRF to forward the packet into after the transport label is popped. This design enables efficient, scalable service delivery.

Operational Benefits

SR-MPLS L3VPN provides several operational advantages. First, it simplifies operations by eliminating the need for LDP troubleshooting and management. Second, it enables faster convergence by reacting immediately to IGP changes. Third, it provides 100% fast-reroute coverage

via TI-LFA, protecting all prefixes without requiring RSVP-TE. Fourth, it scales to thousands of services per PE router, making it ideal for large service providers.

Enterprise Use Cases

Enterprises use SR-MPLS L3VPN for multi-site VPN services, connecting branch offices with guaranteed SLAs. Data center replication benefits from low-latency backup traffic. Hybrid cloud connectivity enables consistent routing between on-premises and cloud environments. Disaster recovery is accelerated by fast failover via TI-LFA.

Service Provider Use Cases

Service providers offer managed L3VPN services to enterprise customers, guaranteeing service level agreements. 5G backhaul uses SR-MPLS L3VPN to transport RAN traffic with low latency and guaranteed bandwidth. SD-WAN underlay provides overlay transport with QoS guarantees. Internet peering benefits from efficient traffic engineering.

Next Steps for Mastery

To deepen expertise in SR-MPLS L3VPN, engineers should study segment routing policies (SR-TE) for advanced traffic engineering. Implementation of TI-LFA for fast reroute provides 100% coverage. Exploration of inter-AS L3VPN (Option C) enables service across multiple autonomous systems. Labbing with EVPN overlay for L2VPN services extends the architecture to Ethernet services.

CONCLUSION

This SR-MPLS L3VPN engineering poster provides a comprehensive technical reference for service provider networks. The lab demonstrates the complete architecture of modern MPLS-based VPN services, from the simplified control plane enabled by Segment Routing to the scalable, multi-tenant service delivery via MP-BGP VPNv4 and VRFs. Engineers completing this lab gain hands-on experience with technologies that are industry standards for modern ISPs, telcos, and large enterprises. The knowledge and skills acquired are directly applicable to CCIE/JNCIE certification exams and to real-world network deployments.

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